

NORRKOPING BRAVALLA, SwedenWMO: **025700**Lat: **58.617N**Lon: **16.117E**Elev: **33**StdP: **100.93**Time Zone: **1.00 (EUC)**Period: **82-93**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.0	-14.5	-21.6	0.5	-18.0	-18.1	0.8	-14.1	13.0	6.4	12.0	4.3	2.5	280	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.3	17.9	24.4	17.0	22.6	16.2	18.9	23.8	17.9	22.8	17.0	21.3	3.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	20.3	16.1	11.5	19.5	15.2	10.8	18.6	53.6	23.8	50.5	22.7	47.9	21.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.0	8.7	DB	-18.6	28.9	5.8	1.9	-22.8	30.2	-26.2	31.3	-29.5	32.3	-33.7	33.7	(4)
(5)			WB	-19.0	20.9	5.8	1.5	-23.1	21.9	-26.5	22.8	-29.8	23.7	-34.0	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-2.2	-2.5	1.0	4.8	11.1	14.5	16.8	15.7	11.2	7.2	2.3	-0.8
	DBStd	7.87	6.18	5.82	3.99	3.46	3.32	3.18	2.44	2.70	2.66	3.27	3.74	4.42
	HDD10.0	1880	379	351	280	160	27	3	0	0	17	96	233	334
	HDD18.3	4297	638	584	538	407	226	122	60	89	213	346	482	592
	CDD10.0	651	0	0	1	3	60	137	211	177	54	8	0	0
	CDD18.3	27	0	0	0	0	0	6	13	8	0	0	0	0
	CDH23.3	276	0	0	0	0	13	58	130	74	1	0	0	0
	CDH26.7	36	0	0	0	0	0	5	18	13	0	0	0	0
Wind	WSAvg	4.0	4.8	4.2	4.2	3.8	4.0	3.4	3.3	3.4	4.0	4.2	4.2	4.2
Precipitation	PrecAvg	611	43	36	30	35	42	68	73	72	58	55	52	48
	PrecMax	795	86	60	57	88	89	125	141	174	129	140	117	88
	PrecMin	438	8	14	8	3	8	13	5	9	11	3	11	13
	PrecStd	73	20	12	14	22	20	27	34	34	33	29	25	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	10.8	13.1	18.9	25.7	27.3	28.6	28.5	22.3	16.8	11.2	9.2
		MCWB	7.4	8.2	8.9	11.4	16.3	18.0	19.0	18.4	17.2	14.1	9.5	7.6
	2%	DB	7.1	8.4	9.7	15.3	23.0	25.0	26.7	25.4	18.6	14.1	10.0	7.5
		MCWB	5.8	6.7	6.5	9.4	14.7	17.0	18.1	17.8	14.2	12.2	8.4	6.2
	5%	DB	5.9	6.4	7.8	12.7	20.4	23.0	24.6	22.8	17.0	12.9	8.6	5.9
		MCWB	4.5	5.0	5.4	8.1	13.0	15.7	17.5	16.7	13.3	10.9	7.2	4.8
	10%	DB	4.6	4.4	6.1	11.0	17.9	20.9	22.6	21.0	15.7	11.8	7.3	4.5
		MCWB	3.4	3.4	4.4	7.2	11.8	14.6	16.7	15.9	12.7	10.0	6.2	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	8.6	9.1	11.9	17.2	19.2	20.8	20.2	17.6	14.5	10.0	7.6
		MCDB	8.9	10.5	12.3	17.9	23.9	24.7	26.4	24.3	20.8	15.6	11.2	8.7
	2%	WB	5.9	6.7	7.1	9.9	15.2	18.0	19.1	18.6	15.6	12.6	8.5	6.4
		MCDB	6.9	8.1	8.8	13.9	21.8	23.2	24.4	23.1	17.4	13.9	9.7	7.4
	5%	WB	4.7	5.1	5.7	8.7	13.7	16.8	18.0	17.5	14.2	11.4	7.4	5.0
		MCDB	5.7	6.2	7.2	12.4	18.9	21.3	22.9	22.0	16.2	12.6	8.4	5.8
	10%	WB	3.5	3.5	4.5	7.5	12.4	15.6	17.2	16.5	13.1	10.3	6.4	3.8
		MCDB	4.5	4.3	5.9	10.3	17.1	20.0	21.5	20.2	15.0	11.5	7.2	4.4
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.5	5.9	8.0	9.9	9.0	9.3	8.6	7.5	5.9	4.4	4.8
		MCDBR	4.7	5.3	8.2	11.8	14.5	12.7	13.4	11.9	9.1	6.9	5.2	5.5
		MCWBR	4.4	4.4	5.5	7.2	7.9	5.9	6.3	5.6	5.5	5.1	4.8	5.2
	5% WB	MCDBR	4.6	5.2	6.9	10.5	12.5	11.2	11.4	10.3	7.9	6.3	4.8	5.4
		MCWBR	4.4	4.6	5.2	6.8	7.4	5.7	6.0	5.6	5.4	5.1	4.6	5.3
Clear-Sky Solar Irradiance	taub	0.284	0.321	0.340	0.378	0.374	0.363	0.390	0.381	0.348	0.338	0.310	0.270	
	taud	2.241	2.295	2.350	2.308	2.385	2.441	2.386	2.408	2.484	2.419	2.289	2.175	
	Ebn at Noon	544	700	802	826	858	874	837	818	792	678	504	423	
	Edh at Noon	45	70	88	109	107	103	107	98	78	63	44	34	
All-Sky Solar Radiation	RadAvg	0.34	0.93	2.44	3.98	5.23	5.80	5.26	4.18	2.81	1.26	0.43	0.23	
	RadStd	0.05	0.16	0.26	0.41	0.43	0.27	0.53	0.41	0.35	0.20	0.09	0.03	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (55 neighbors)	+0.40	N/A	N/A	N/A	N/A	-128	-159	N/A	N/A	(47)

Nomenclature: See separate page